$$\underline{f}_p = \frac{k(a-a_0)}{e_p} \pm \times \underline{n}_p = \frac{k(a-a_0)}{e_p}\left(\underline{n}_u - (\underline{n}_p^T \underline{n}_u)\,\underline{n}_p\right);$$

$$\frac{\partial \underline{f}_p}{\partial \alpha} = k(a-a_0)\left[\frac{\partial}{\partial \alpha}\left(\frac{\underline{n}_u}{e_p}\right) - \frac{\partial}{\partial \alpha}\left(\underline{n}_p^T \underline{n}_u\right)\frac{\underline{n}_p}{e_p}\right] =$$

$(=)$

$$\frac{\partial e_p}{\partial \alpha} = \underline{n}_p^T \frac{\partial \underline{p}_e}{\partial \alpha}; \quad \frac{\partial \underline{n}_p}{\partial \alpha} = \frac{1}{e_p}\left[\underline{\underline{N}}_{pp} - \underline{\underline{I}}\right]\frac{\partial \underline{p}_e}{\partial \alpha};$$

$$\frac{\partial}{\partial \alpha}\left(\underline{n}_p^T \underline{n}_u\right) = \underline{n}_p^T \cdot \frac{\partial \underline{n}_u}{\partial \alpha} + \underline{n}_u^T \cdot \frac{\partial \underline{n}_p}{\partial \alpha};$$

$$\frac{\partial}{\partial \alpha}\left[\left(\underline{n}_p^T \underline{n}_u\right)\underline{n}_p\right] = \underline{\underline{N}}_{pp} \frac{\partial \underline{n}_u}{\partial \alpha} + \underline{\underline{N}}_{pu} \frac{\partial \underline{n}_p}{\partial \alpha} = \text{(not forgot to div by } e_p\text{.)}$$

$$=, \quad \frac{\underline{\underline{N}}_{pp}}{e_u}\left[\underline{\underline{I}} - \underline{\underline{N}}_{uu}\right]\frac{\partial \underline{p}_e}{\partial \alpha} + \frac{\underline{\underline{N}}_{pu}}{e_p}\left[\underline{\underline{I}} - \underline{\underline{N}}_{pp}\right]\frac{\partial \underline{p}_e}{\partial \alpha};$$

$$\frac{\partial}{\partial \alpha}\left(\frac{\underline{n}_u}{e_p}\right) = \frac{1}{e_p^2}\left(e_p \frac{\partial \underline{n}_u}{\partial \alpha} - \underline{n}_u \cdot \frac{\partial e_p}{\partial \alpha}\right) = \frac{1}{e_p e_u}\left[\underline{\underline{I}} - \underline{\underline{N}}_{uu}\right]\frac{\partial \underline{p}_e}{\partial \alpha} -$$

$$- \frac{\underline{\underline{N}}_{up}}{e_p^2}\frac{\partial \underline{p}_e}{\partial \alpha};$$

$(=)$

$$k(a-a_0)\cdot\left[\frac{\underline{\underline{N}}_{pp}}{e_p e_u}\cdot\frac{\partial \underline{p}_e}{\partial \alpha} - \frac{\underline{\underline{N}}_{up}}{e_p^2}\cdot\frac{\partial \underline{p}_e}{\partial \alpha} - \frac{\underline{\underline{N}}_{pp}}{e_u e_p}\left[\underline{\underline{I}} - \underline{\underline{N}}_{uu}\right]\frac{\partial \underline{p}_e}{\partial \alpha}\right.$$

$$\left. - \frac{\underline{\underline{N}}_{pu}}{e_p^2}\left[\underline{\underline{I}} - \underline{\underline{N}}_{pp}\right]\frac{\partial \underline{p}_e}{\partial \alpha} - \frac{\underline{n}_p^T \underline{n}_u}{e_p^2}\cdot\left[\underline{\underline{I}} - \underline{\underline{N}}_{pp}\right]\frac{\partial \underline{p}_e}{\partial \alpha} + \frac{(\underline{n}_p^T \underline{n}_u)\cdot \underline{\underline{N}}_{pp}}{e_p^2}\frac{\partial \underline{p}_e}{\partial \alpha}\right]$$

$$\underline{f}_n = \frac{k(a-a_0)}{e_n}\underline{n}_n \times \underline{m}_n = \frac{k(a-a_0)}{e_p}\left((\underline{n}_p^T \underline{n}_n)\underline{n}_n - \underline{n}_p\right)$$

$$\frac{\partial \underline{f}_n}{\partial \underline{\alpha}} = k(a-a_0)\left[\frac{\partial}{\partial \underline{\alpha}}(\underline{n}_p^T \underline{n}_n)\frac{\underline{n}_n}{e_n} - \frac{\underline{n}_p}{e_n}\right];$$

$$(=)$$

$$\frac{\partial e_p}{\partial \underline{\alpha}} = \underline{n}_p^T \frac{\partial \underline{p}_e}{\partial \underline{\alpha}}; \quad \frac{\partial \underline{n}_p}{\partial \underline{\alpha}} = \frac{1}{e_p}\left[\underline{\underline{I}} - \underline{n}_p \underline{n}_p^T\right]\frac{\partial \underline{p}_e}{\partial \underline{\alpha}}; \quad \left[\underline{\underline{I}} - \underline{n}_p \underline{n}_p^T\right] = \underline{\underline{N}}_p$$

$$\frac{\partial}{\partial \underline{\alpha}}(\underline{n}_p^T \underline{n}_n) = \underline{n}_p^T \frac{\partial \underline{n}_n}{\partial \underline{\alpha}} + \underline{n}_n^T \frac{\partial \underline{n}_p}{\partial \underline{\alpha}};$$

$$\frac{\partial}{\partial \underline{\alpha}}\left[(\underline{n}_p^T \underline{n}_n)\underline{n}_n\right] = \underline{\underline{N}}_p \frac{\partial \underline{n}_n}{\partial \underline{p}_e}\frac{\partial \underline{p}_e}{\partial \underline{\alpha}} + \underline{n}_n \underline{n}_n^T \frac{\partial \underline{n}_p}{\partial \underline{p}_e}\frac{\partial \underline{p}_e}{\partial \underline{\alpha}} = \text{(not needed to dif by } \underline{p}_e\text{)}$$

$$= \frac{\underline{\underline{N}}_p}{e_n}\left[\underline{\underline{N}}_n \underline{\underline{N}}_n - \underline{\underline{I}}\right]\frac{\partial \underline{p}_e}{\partial \underline{\alpha}} + \frac{\underline{n}_n \underline{n}_n^T}{e_p}\left[\underline{\underline{I}} - \underline{\underline{N}}_p\right]\frac{\partial \underline{p}_e}{\partial \underline{\alpha}};$$

$$\frac{\partial}{\partial \underline{\alpha}}\left(\frac{\underline{n}_p}{e_n}\right) = \frac{1}{e_n^2}\left(e_n \frac{\partial \underline{n}_p}{\partial \underline{p}_e}\frac{\partial \underline{p}_e}{\partial \underline{\alpha}} - \underline{n}_p \frac{\partial e_n}{\partial \underline{p}_e}\frac{\partial \underline{p}_e}{\partial \underline{\alpha}}\right) = \frac{1}{e_p e_n}\left[\underline{\underline{I}} - \underline{n}_p \underline{n}_p^T\right]\frac{\partial \underline{p}_e}{\partial \underline{\alpha}} -$$

$$- \frac{\underline{\underline{N}}_p \underline{n}_n}{e_n^2}\frac{\partial \underline{p}_e}{\partial \underline{\alpha}}$$

$$(=) \; k(a-a_0)\left[\frac{\underline{\underline{N}}_p}{e_n^2}\left[\underline{\underline{I}} - \underline{n}_n \underline{n}_n^T\right]\frac{\partial \underline{p}_e}{\partial \underline{\alpha}} + \frac{\underline{\underline{N}}_n}{e_p e_n}\left[\underline{n}_n \underline{n}_p^T - (\underline{n}_p^T \underline{n}_n)\underline{\underline{I}}\right]\frac{\partial \underline{p}_e}{\partial \underline{p}_e} + \frac{\underline{n}_p^T \underline{n}_n}{e_n^2}\left[\underline{\underline{I}} - \underline{\underline{N}}_n\right]\frac{\partial \underline{p}_e}{\partial \underline{p}_e}\right.$$

$$\left. - \frac{(\underline{n}_p^T \underline{n}_n)}{e_n^2}\underline{\underline{N}}_n \frac{\partial \underline{p}_e}{\partial \underline{\alpha}} - \frac{1}{e_p e_n}\left[\underline{\underline{I}} - \underline{n}_p \underline{n}_p^T\right]\frac{\partial \underline{p}_e}{\partial \underline{p}_e} + \frac{\underline{\underline{N}}_p}{e_n^2}\frac{\partial \underline{p}_e}{\partial \underline{\alpha}}\right].$$

$$\underline{\underline{J}}_p = k(a-a_0)\left[\frac{1}{e_p e_n}\cdot(\underline{\underline{I}}-\underline{\underline{N}}_{pp})\cdot(\underline{\underline{I}}-\underline{\underline{N}}_{nn})\frac{\partial \underline{d}_n}{\partial \underline{d}} + \right.$$

$$\left. + \left\{ \frac{(\mathcal{A}\cdot\underline{\underline{N}}_{pp} - \underline{\underline{N}}_{np})}{e_p^2} - \frac{(\underline{\underline{N}}_{pn} + \mathcal{A}\cdot\underline{\underline{I}})}{e_p^2}\cdot(\underline{\underline{I}}-\underline{\underline{N}}_{pp})\right\}\frac{\partial \underline{d}_p}{\partial \underline{d}}\right]$$

$$\underline{\underline{J}}_n = k(a-a_0)\left[\frac{1}{e_n^2}\left\{(\underline{\underline{N}}_{pn}+\mathcal{A}\cdot\underline{\underline{I}})\cdot(\underline{\underline{I}}-\underline{\underline{N}}_{nn}) + (\underline{\underline{N}}_{pn}-\mathcal{A}\cdot\underline{\underline{N}}_{nn})\right\}\frac{\partial \underline{d}_n}{\partial \underline{d}} + \right.$$

$$\left. + \frac{1}{e_n e_p}(\underline{\underline{N}}_{nn}-\underline{\underline{I}})\cdot(\underline{\underline{I}}-\underline{\underline{N}}_{pp})\frac{\partial \underline{d}_p}{\partial \underline{d}}\right];$$

$$\mathcal{A} = \underline{n}_p^T \cdot \underline{n}_n$$

Radu's notation $\quad \underline{n} \to \underline{d}$

$$\underline{d} \to \underline{v}$$

$$\underline{\underline{N}} \to \underline{\underline{D}}$$

$$\underline{\underline{J}}_p = k(a-a_0)\left[\frac{\underline{\underline{I}}-\underline{\underline{D}}_{pp}}{e_p e_n}\cdot(\underline{\underline{I}}-\underline{\underline{D}}_{nn})\frac{\partial \underline{v}_n}{\partial \underline{d}} + \frac{1}{e_p^2}\left\{(\mathcal{A}\cdot\underline{\underline{D}}_{pp}-\underline{\underline{D}}_{np})-(\underline{\underline{D}}_{pn}+\mathcal{A}\cdot\underline{\underline{I}})\cdot\right.\right.$$

$$\left.\left.\cdot(\underline{\underline{I}}-\underline{\underline{D}}_{pp})\right\}\frac{\partial \underline{v}_d}{\partial \underline{d}}\right];$$

$$\underline{\underline{J}}_n = k(a-a_0)\left[\frac{1}{e_n^2}\left\{(\underline{\underline{D}}_{pn}+\mathcal{A}\cdot\underline{\underline{I}})(\underline{\underline{I}}-\underline{\underline{D}}_{nn}) + (\underline{\underline{D}}_{pn}-\mathcal{A}\cdot\underline{\underline{D}}_{nn})\right\}\frac{\partial \underline{v}_n}{\partial \underline{d}} + \frac{1}{e_n e_n}\cdot\right.$$

$$\left.\cdot(\underline{\underline{D}}_{nn}-\underline{\underline{I}})\cdot(\underline{\underline{I}}-\underline{\underline{D}}_{pp})\frac{\partial \underline{v}_d}{\partial \underline{d}}\right];$$